PROJECT 67 · REVIEW 3

DBMS REVIEW-3

Gym Membership and Fitness Session Management System

Application Development, Testing and Final Demonstration

Student Names · Roll Numbers
Faculty Guide · Department · University



From schema to working system

Review 3 closes the loop: connected application, controlled operations and demonstrable evidence.

CENTRALIZED SCOPE



Members · subscriptions · trainers · classes · attendance

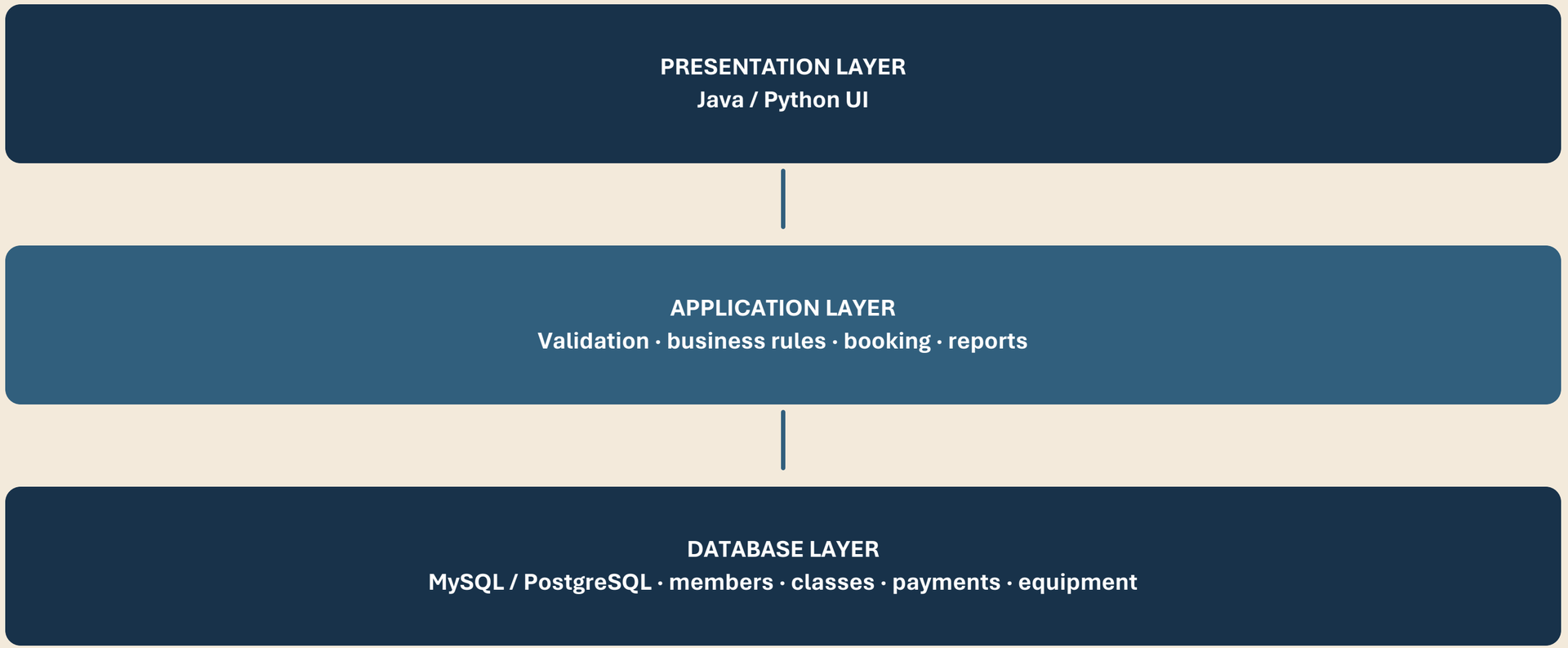
Assessments · workout plans · schedules · payments · equipment · maintenance

Review 1 — requirements, users, scope, ER model **Review 2 — 3NF, constraints, SQL** **Review 3 — application, validation, reports, testing**

Three layers. One trusted data path.

Separation of concerns keeps the interface responsive, rules consistent and records reliable.

PKs, FKs and constraints preserve integrity



USER → APPLICATION → DATABASE → QUERY RESULTS → USER INTERFACE

Technology stack & connectivity

A desktop client communicates through parameterized queries and a controlled transaction lifecycle.

FRONTEND

Java Swing / JavaFX
or Python Tkinter

BACKEND

CRUD · validation
business logic

DATABASE

MySQL / PostgreSQL

CONNECTOR

JDBC / Python DB connector



Credentials secured · graceful connection and query error handling

One application. Role-aware views.

The dashboard reveals only the actions and records each role is authorized to access.

ADMINISTRATOR

Members · plans · trainers
Schedules · equipment · reports

MEMBER MANAGEMENT

Membership · classes
Bookings · renewal status

TRAINER PORTAL

Assigned sessions
Bookings · attendance

MANAGEMENT REPORTS

Subscriptions · attendance
Revenue · maintenance

CRUD OPERATIONS

1 → 2 → 3 → 4

CREATE

Add members, trainers and classes

READ

View and search records

UPDATE

Edit member details and renew memberships

DELETE

Remove permitted records while maintaining data integrity.

EXAMPLE SQL

```
INSERT INTO Members VALUES
(101, 'Rahul', 'rahul@example.com', '9876543210');

SELECT * FROM Members;

UPDATE Members SET Name = 'Rahul K'
WHERE Member_ID = 101;

DELETE FROM Members WHERE Member_ID = 101;
```



Benefit:

Simplifies record management and maintains data consistency.

Member & membership management

A single workspace supports registration, search, plan assignment and lifecycle visibility.

MEMBER FORM

- Member ID
- Name
- Email
- Phone
- Plan
- Start Date
- End Date
- Subscription Status

MEMBERS

ID	NAME	PLAN	END DATE	STATUS
M-104	A. Sharma	Annual	14 Jun 2026	ACTIVE
M-105	N. Rao	Monthly	22 Mar 2026	EXPIRING
M-106	J. Thomas	Quarterly	02 Feb 2026	EXPIRED

CREATE → READ → UPDATE → DELETE

Illustrative application screen — replace with genuine build screenshot

Trainer assignment & fitness records

Validated assessments and conflict-aware scheduling connect people, programmes and progress.

TRAINER ASSIGNMENT

Profiles & specializations

Contact information

Assigned classes and sessions

Schedule overlap detection

Illustrative interface placeholder

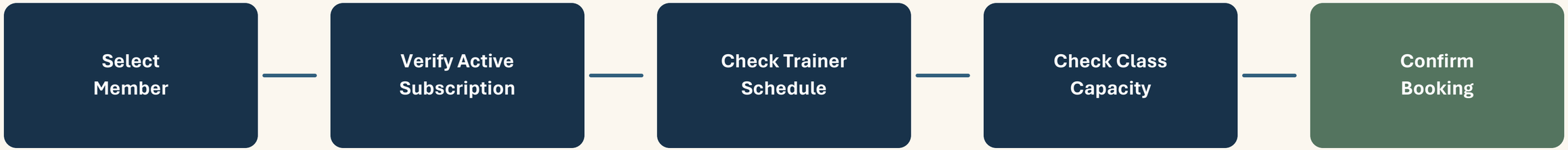
ASSESSMENT / PROGRAMME

Weight	72.5 kg
Height	178 cm
Body-fat	18.2%
Assigned trainer	[select]
Programme dates	[from — to]

VALIDATE: no negative values · valid dates · no trainer overlap

Booking is a controlled transaction

A seat is confirmed only after subscription, trainer and capacity checks all pass.



CAPACITY RULE

```
confirmed registrations < maximum capacity
```

CONCURRENCY CONTROL

transaction + database locking
prevents double booking

Any failed rule → reject safely and roll back

Attendance becomes an operational signal

Authorized trainers and administrators record status once; summaries serve multiple views.

REG ID	MEMBER	CLASS	SCHEDULE	STATUS
R-201	A. Sharma	Strength	08:00–09:00	ATTENDED
R-202	N. Rao	Yoga	09:30–10:30	ABSENT
R-203	J. Thomas	Cardio	18:00–19:00	ATTENDED
R-204	P. Sen	Strength	08:00–09:00	REGISTERED

Illustrative attendance table

SUMMARY

24
registered

18
attended

6
absent

Payments, renewals & expiry tracking

Database-led status reflects actual outcomes; renewal history preserves both old and new dates.

PAYMENT HISTORY

DATE	AMOUNT	METHOD	STATUS
04 Mar 26	2,500	Card	PAID
04 Feb 26	2,500	Transfer	FAILED
04 Jan 26	2,500	Cash	PAID

Outstanding and failed payments remain visible — no dummy success claims.

RENEW MEMBERSHIP

Member ID	M-105
Old expiry	22 Mar 2026
New expiry	22 Apr 2026
Payment status	Confirmed

Expired · expiring soon · renewed

Equipment availability, planned—not guessed

Maintenance history and next-service dates turn records into safer operating decisions.

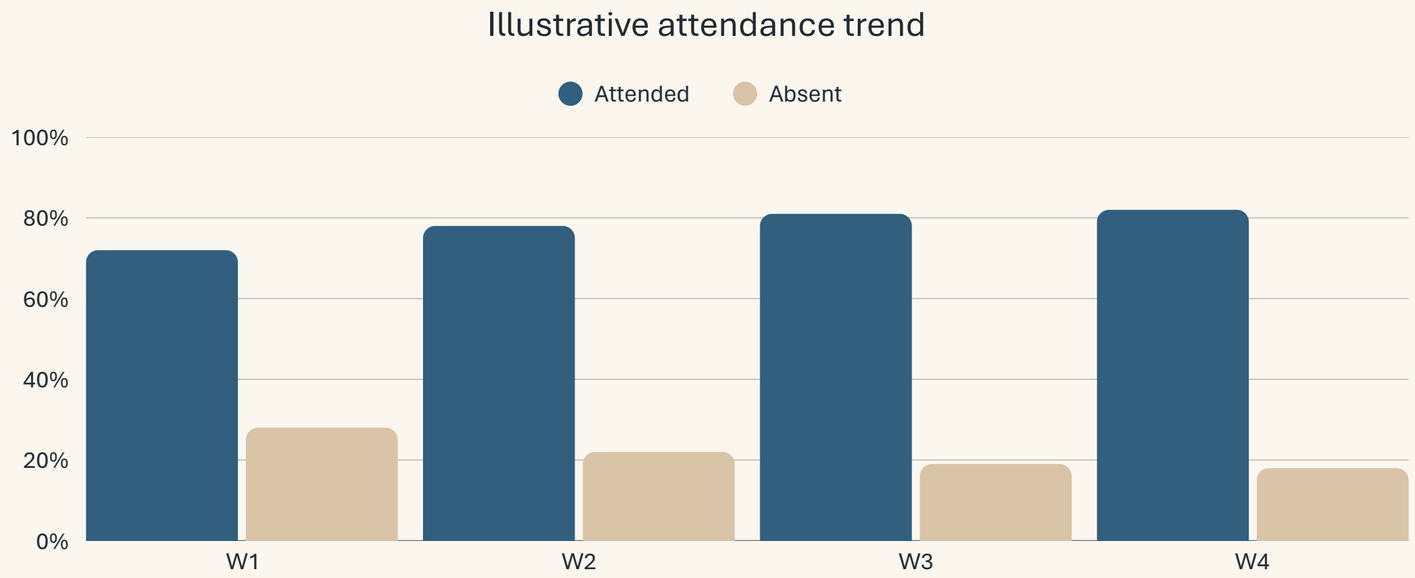
EQUIPMENT ID	NAME	STATUS	LAST MAINT.	NEXT MAINT.	COST
EQ-011	Treadmill 3	AVAILABLE	12 Jan 2026	12 Apr 2026	4,500
EQ-018	Cable Station	MAINTENANCE	28 Feb 2026	15 Mar 2026	7,800
EQ-024	Spin Bike 6	UNAVAILABLE	04 Mar 2026	10 Mar 2026	2,200

Service planning → fewer disruptions → higher equipment availability

SQL views power decision-ready reports

Illustrative dashboard values below must be replaced by actual sample-data query results.

128	14	82%	₹—
Active memberships	Upcoming renewals	Attendance rate	Revenue



MEMBERSHIP
active · expired · renewal

TRAINER
sessions · assigned classes

FINANCIAL + EQUIPMENT
revenue · maintenance cost